Discussion and Conclusion

Discussion and Conclusion I

Numbered-table references point to the canonical manuscript; they are not standalone local deck numbers.

DuckRabbit's principal result is a reproducibility boundary rather than a new psychophysical finding. The same typed request can be regenerated, hashed, encoded, decoded, and audited, while the evidence graph and observer layer remain explicit about what the package cannot infer. This makes the atlas useful for stimulus construction, source comparison, and preregistration without treating a rendered image or sound as behavioral evidence.

The scholarly contribution is correspondingly modest but operational. Rather than flattening visual, auditory, temporal, and multisensory families into a single "illusion" label, the package keeps mechanism, perceptual signature, requirements, evidence role, engineering basis, and implementation status separate. A primary demonstration, a review, a theoretical account, and a source describing the DuckRabbit implementation answer different questions. The appendix and source-data sidecars make those distinctions inspectable at the same time as the generated media.

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The work is best understood as a research-software artifact with a bounded methods contribution. Its novelty claim is not that it discovers a new illusion or resolves a disputed mechanism; it is that a heterogeneous stimulus catalog can be represented as typed, reproducible, source-linked, and release-audited objects. This framing is consistent with software-citation and FAIR guidance, which treats versioned software, metadata, provenance, and reuse conditions as part of the research record (Smith et al. 2016; Wilkinson et al. 2016; Lamprecht et al. 2020).

The release is authored by Daniel Ari Friedman of the Active Inference Institute and is published and citable at the DuckRabbit public GitHub repository. The v0.5.0 archive is a published versioned record. Subsequent source and manuscript corrections remain unreleased until a new version is explicitly tagged and deposited. A local package audit does not certify that the archived PDF was rebuilt from the current development manuscript.

Conclusion I

DuckRabbit v0.6.0 turns multimodal illusion generation into an auditable typed pipeline. A request produces a canonical artifact, objective metrics, optional delivery files, decoded inspection, and a versioned evidence-bounded manifest. The catalog now records not only what is implemented, but also what the literature supports and what remains unvalidated.

The main scientific contribution is a boundary: deterministic stimulus facts are reproducible software outputs, while perceptual effects are hypotheses that require observer conditions and data. This boundary permits broad generator coverage without overclaiming. Future additions should contribute a typed parameter contract, a literature record, an engineering-fidelity statement, objective invariants, generated documentation, and tests before entering the implemented registry. Promotion is a release decision about software readiness, not a verdict

on whether a perceptual phenomenon is real.

The synthetic-psychophysics layer extends this boundary without crossing it. A transparent feature observer makes model inputs, weights, calibration, and output hashes inspectable, so the full orchestration can be tested today. Its small curve is a diagnostic of that specified model, not evidence that humans share its feature map or response function. Empirical observer work remains a separate, ethically and methodologically governed stage.